

A Comparative Study of BVARs with Flexible Innovation Distributions

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Outline

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- Bayesian Vector Autoregressions (BVARs) are widely used for modeling multivariate time series (financial forecasting).
- Standard BVARs assume Gaussian innovations (limitations).
- Financial returns exhibit:
 - Heavy tails
 - Volatility clustering
 - Structural changes
- This leads to **model misspecification** under Gaussian assumptions.
- Goal: Evaluate more flexible innovation distributions under the VAR framework.

Motivation

- Gaussian VAR models are simple and convenient.
- Underestimate the probability of extreme losses.
- This becomes severe during periods of market stress (COVID-19 and the Global Financial Crisis).
- More flexible innovation structures may improve:
 - Predictive accuracy
 - Density forecasting
 - Tail-risk estimation
- Main question: Can non-Gaussian innovations better capture financial return dynamics?



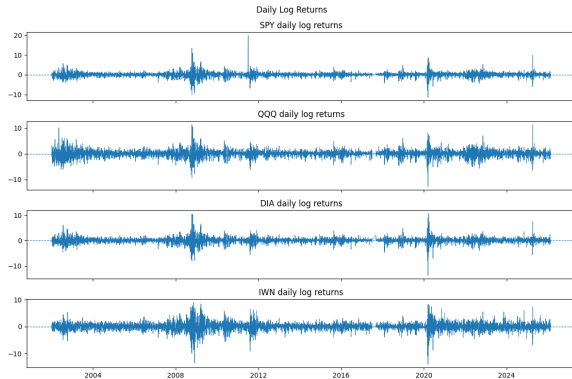
- VAR models are a standard framework for multivariate time-series analysis.
- But unrestricted VARs become overparameterized in larger systems.
- BVARs solve this problem through shrinkage priors, especially the Minnesota prior.
- Gaussian innovations are often too restrictive for financial returns.
- The literature mainly extends BVARs in three directions:
 - **Student-t** for heavy tails,
 - **Mixture** for regime changes,
 - **Stochastic volatility** for volatility clustering.

Research Gap and Our Contribution

- Existing studies often examine flexible innovation models separately.
- A unified comparison is useful for understanding their relative strengths.
- This paper compares four models:
 - Gaussian VAR
 - Student- t VAR
 - Mixture VAR
 - Stochastic Volatility VAR
- The comparison focuses on:
 - Predictive density accuracy
 - Value-at-Risk (VaR)
 - Expected Shortfall (ES)

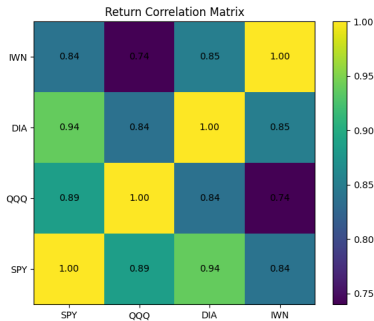
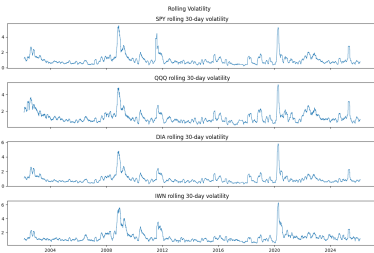
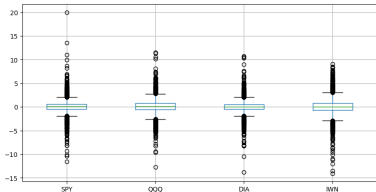
Data Description

- Daily log returns for four ETFs:
 - SPY (S&P 500)
 - QQQ (NASDAQ 100)
 - DIA (Dow Jones)
 - IWN (Russell 2000 Value)
- Sample: 2002 – 2026 (~6000 observations)
- Covers major market regimes:
 - 2008 financial crisis
 - COVID-19 shock



Stylized Facts of Returns

- **Heavy tails:** Extreme observations beyond Gaussian expectations
- **Volatility clustering:** Persistent periods of high/low variance
- **Strong correlation:** High co-movement across assets



Modeling Framework

- We consider a VAR(p) model with flexible innovations:

$$y_t = Bx_t + \epsilon_t$$

- The key modeling choice lies in the distribution of ϵ_t :
 - **Gaussian**: baseline assumption
 - **Student- t** : heavy-tailed innovations
 - **Mixture**: regime-dependent volatility
 - **SV**: time-varying volatility
- Key idea:

Instead of changing the mean dynamics, we improve the model by making the innovation distribution more realistic.

Non-Gaussian Innovations

- We consider two flexible alternatives to Gaussian innovations:

- **Student- t innovations:**

- heavy-tailed distribution
- equivalent to a scale mixture:

$$\epsilon_t \mid \lambda_t \sim \mathcal{N}(0, \Sigma / \lambda_t)$$

- **Mixture innovations:**

- mixture of Gaussians:

$$\epsilon_t \sim \sum_{k=1}^K \pi_k \mathcal{N}(0, \Sigma_k)$$

- captures regime switching
- Key idea:

Student- t handles outliers, while mixture models capture regime changes.

Two Estimation Paradigms

- We study the same VAR models under two estimation approaches:

- **Full Bayesian Inference:**

- infer posterior distribution:

$$p(\theta \mid Y)$$

- accounts for parameter uncertainty
 - predictive distribution integrates over posterior

- **Point Estimation:**

- estimate parameters via MLE / EM
 - obtain a single estimate $\hat{\theta}$
 - predictive distribution conditional on $\hat{\theta}$

- Key question:

Are performance gains driven by better innovation modeling, or by Bayesian inference itself?

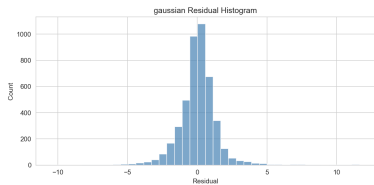
Point Estimation

- We use point estimation as a controlled benchmark.
- Goal:
 - isolate the effect of innovation distributions
 - remove the influence of posterior uncertainty
- Estimation methods:
 - **Gaussian VAR:** OLS / maximum likelihood
 - **Student- t VAR:** maximum likelihood
 - **Mixture VAR:** EM algorithm
- Interpretation:

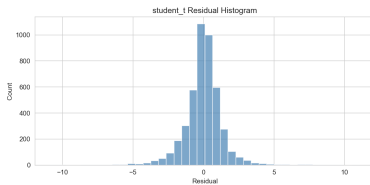
If non-Gaussian models still outperform under point estimation, the improvement comes from better modeling of innovations rather than Bayesian inference.

Point Estimation Results: Residual Diagnostics

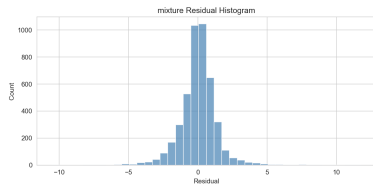
- The main differences across models appear in the residual distribution rather than in the estimated mean dynamics.



Gaussian



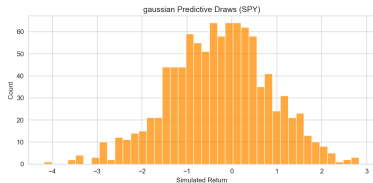
Student-t



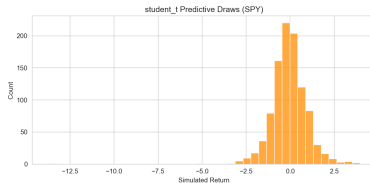
Mixture

Point Estimation Results: Predictive Distributions

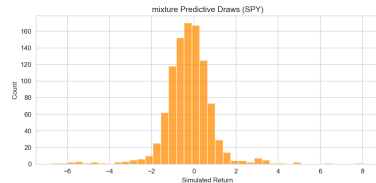
- We next compare the predictive distributions implied by point estimation.
- **Gaussian VAR:**
 - produces relatively thin-tailed predictive densities
 - tends to underestimate the probability of extreme outcomes
- **Student- t VAR:**
 - generates heavier tails through a more robust innovation distribution
 - better reflects uncertainty during volatile periods
- **Mixture VAR:**
 - allows more flexible predictive shapes
 - reflects multiple volatility regimes in the data



Gaussian



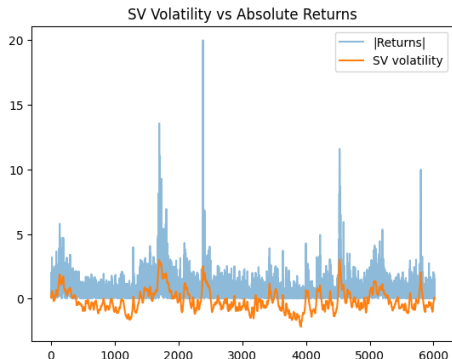
Student- t



Mixture

Extensions: Stochastic Volatility VAR (SV-VAR)

- Time-varying variance model: $\epsilon_t \sim \mathcal{N}(0, \text{diag}(\exp(h_t)))$
- Latent volatility follows AR(1): $h_t = \phi h_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_h^2)$
- Key idea: Captures volatility clustering and models continuous changes in risk
- Limitations: Does not fully capture heavy tails, and is less effective for extreme events



MCMC Estimation

- Posterior inference via Gibbs sampling:

$$p(\theta \mid Y), \quad \theta = (B, \Sigma, \text{latent variables})$$

- Iterative updates (conditional conjugacy):

- 1 Sample latent variables:

$$\lambda_t \text{ (Student-}t\text{)}, \quad z_t \text{ (Mixture)}, \quad h_t \text{ (SV)}$$

- 2 Sample regression coefficients:

$$B \mid \Sigma, \text{latent}, Y$$

- 3 Sample covariance parameters:

$$\Sigma \mid B, \text{latent}, Y$$

- Predictive sampling:

$$y_{t+1}^{(s)} \sim p(y_{t+1} \mid \theta^{(s)})$$

- Diagnostics:

- Effective Sample Size (ESS)
- Trace plots and stability

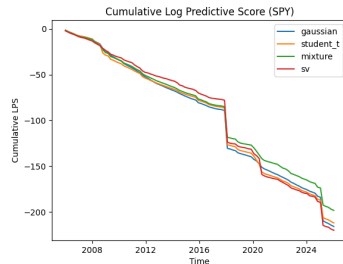
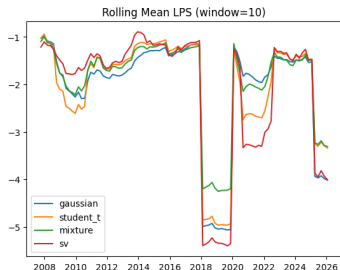
Out-of-Sample Performance

- Rolling one-step-ahead forecasts for SPY
- Metrics:
 - Log Predictive Score (higher is better)
 - Expected Shortfall (more negative is better)
- Key findings:
 - Mixture VAR achieves best predictive accuracy
 - Best tail risk estimation (ES)
 - Gaussian provides unreliable tail risk estimates

Model	Mean LPS	VaR _{5%}	ES _{5%}	Avg. Pred. Std
Mixture	-1.9656	-1.7524	-2.8140	1.1608
Student- <i>t</i>	-2.1010	-1.5624	-2.1849	0.9537
Gaussian	-2.1400	-1.8480	-2.3369	1.1775
SV	-2.1809	-1.3686	-1.7726	0.9531

Time-Varying Predictive Performance

- Evaluate performance over time using:
 - Rolling LPS (short-term performance)
 - Cumulative LPS (overall performance)
- Observations:
 - Performance deteriorates during market stress
 - Flexible models are more stable
 - Mixture VAR consistently outperforms
 - Differences are most pronounced during crisis periods



Model Comparison Summary

Model	Main Strength	Main Limitation
Gaussian VAR	Simple, interpretable	Misses heavy tails
Student- t VAR	Robust to outliers	No regime structure
Mixture VAR	Best predictive fit, tail risk	More complex inference
SV-VAR	Captures volatility clustering	Weaker tail modeling

Main takeaway

Relaxing Gaussian assumptions materially improves probabilistic forecasting and risk estimation in financial time series.

Conclusion

- Gaussian assumptions fail to capture financial return behavior
- Flexible models improve:
 - Predictive accuracy
 - Tail risk estimation
- Key results:
 - Student- t improves robustness
 - Mixture VAR performs best overall
 - SV-VAR captures volatility but misses tails
- **Final message:**

Flexible innovation modeling is essential for realistic financial forecasting and risk estimation.

Limitations and Future Work

- **Limitations:**

- Linear VAR structure
- Fixed number of regimes
- Computationally expensive MCMC for rolling estimation
- Limited data (4 ETFs, no macro variables)

- **Future directions:**

- Data-driven regime selection
- Multivariate volatility modeling
- Hybrid models (tails + SV + regimes)
- Incorporating richer datasets

More details: <https://github.com/JonathanMa03/pmlproject-MaZhuTao-SP26>